

Lecture 03

Programming Fundamentals

Chapter lecture — 13 topics with explanations, worked examples, Urdu notes, practice questions and a full answer key.

BIEK Computer Science XI · Bilingual support: English + اردو

بی آئی ای کے کمپیوٹر سائنس — گیارہویں جماعت

13 book lectures in this pack

★ Golden topics, Urdu notes, review and answer key

XI

How to use this BIEK Computer Science lecture

Each PDF is a classroom lecture from the BIEK / Sindh Computer Science textbooks (New Sindh Curriculum, STBB Grade XI–XII). Read *Learn it*, study the SVG diagram, try the worked example on paper, then answer *Check yourself* before looking at the answers. **Extra board practice** (MCQs, short and long questions) is added after the book content — it is not in the textbook. Career-only extras that were not in the book lecture list have been removed. Math is rendered with MathJax. ★ Golden topics are high-yield in board exams. The green box repeats the idea in Urdu.

ہر لیکچر بی آئی کے / سندھ نصاب کی کتاب کے درس کے مطابق ہے۔ سبق پڑھیں، مثال حل کریں، پھر چیک یورسیلف کے سوالات کے جواب اپنے نوٹ بک میں لکھیں۔ کتاب کے بعد اضافی بورڈ پریکٹس ہے۔ ★ گولڈن ٹاپکس امتحان میں زیادہ پوچھے جاتے ہیں۔

Lectures in this pack

Chapter 3: Programming Fundamentals. Study one lecture at a time, then use the chapter review and answer key at the end of this PDF.

L01. 3.1 Programming & Language Types

L02. 3.2 Career Growth with Python

L03. 3.3 Integrated Development Environment (IDE) & VS Code

L04. 3.4 Fundamentals of Python Programming

L05. 3.4.1 Variables and Data Types

L06. 3.4.2 Input Output Handling

L07. 3.4.3 Operators and Operands (Arithmetic, Relational, Bitwise)

L08. 3.5 Control Structures Overview

L09. 3.5.1 Sequence

L10. 3.5.2 Selection Statements (if, if-else, if-elif-else, nested)

L11. 3.5.3 Repetition Statements (for loop, while loop)

L12. 3.6 Libraries in Python (Built-in & Third-Party)

L13. 3.7 Debugging Code in Python

Programming Fundamentals

In this chapter you start writing real programs in Python. You will learn how programs store data, take input, make decisions, repeat work, use ready-made libraries, and how to find and fix your own mistakes.

اس باب میں آپ پائتھن (Python) میں حقیقی پروگرام لکھنا سیکھیں گے۔ آپ جانیں گے کہ پروگرام ڈیٹا کیسے محفوظ کرتا ہے، فیصلے کیسے کرتا ہے، کام کو کیسے دہراتا ہے، اور غلطیاں کیسے تلاش کی جاتی ہیں۔

What you will learn

- Define a program and a programming language, and compare low-, mid- and high-level languages.
- Describe the features of an IDE and the main parts of the VS Code window.
- Take input with `input()`, convert it with `int()` / `float()`, and display output with `print()` and f-strings.
- Use sequence, selection (`if`, `if-else`, `if-elif-else`, nested `if`) and repetition (`for`, `while`, `range()`, `break`, `continue`, nested loops).
- Identify syntax, runtime and logical errors and read a traceback to fix a bug.
- Name career areas where Python is used.
- Write Python programs that use comments, correct indentation, valid variable names and the main data types.
- Evaluate expressions with arithmetic, assignment, relational, logical, membership and bitwise operators, including binary working.
- Import and use built-in libraries (`math`, `random`, `datetime`) and install third-party libraries with `pip`.

3.1 Programming & Language Types

LEARN IT

Computers are "dumb" machines. They are very fast, but they cannot think or do anything on their own. They need exact, step-by-step **instructions**.

- A set of instructions given to a computer is called a **program**.
- The language used to write these instructions is called a **programming language**.
- Writing programs is called **programming**, and the person who writes them is a **programmer**.

Think of a recipe. A cook follows the recipe line by line: "boil water, add rice, wait 15 minutes". A computer follows a program in the same way, one instruction at a time, without guessing what you meant.

WORKED EXAMPLE

A tiny program. Here is a complete Python program with one instruction. It tells the computer to show a message on the screen.

```
print("Hello")
```

OUTPUT
Hello

The word `print` is the instruction; `"Hello"` is the data it works on.

REMEMBER IT

Program = recipe, programmer = recipe writer, computer = cook who follows every line exactly. If the recipe is wrong, the food is wrong. The cook does not fix it for you.

COMMON MISTAKE

Students often think the computer "understands" what they want. It does not. If you type `Print("Hello")` with a capital P, Python stops with an error, because it only knows `print`. Be exact.

اردو میں سمجھیں

کمپیوٹر خود سے کچھ نہیں سوچ سکتا۔ اسے کام کرنے کے لیے ہدایات چاہئیں۔ ہدایات کے مجموعے کو پروگرام (Program) کہتے ہیں، اور جس زبان میں یہ ہدایات لکھی جاتی ہیں اسے پروگرامنگ لینگویج (Programming Language) کہتے ہیں۔

CHECK YOURSELF

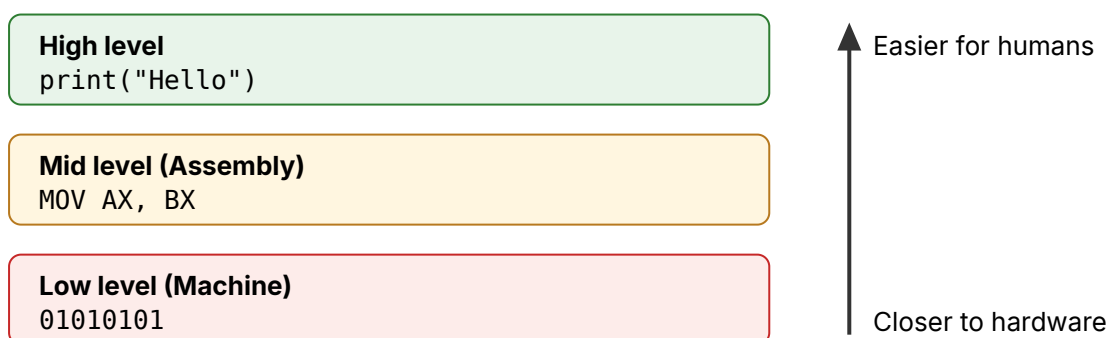
1. What is a program?
2. Why do we say computers are "dumb" machines?

Answers: 1. A set of instructions given to a computer to perform a task. 2. Because they cannot think or do anything without instructions; they only follow exactly what they are told.

LEARN IT

Programming languages are classified into **three main types**, depending on how close they are to the computer's own language (binary 0s and 1s) or to human language.

Type	Description	Examples
Low level	Closest to the native language of the computer (binary 0s and 1s). Very difficult for humans to read and write.	Machine language (binary)
Mid level	Bridges the gap. It can access hardware but is easier to read than binary because it uses short words called mnemonics (e.g. <code>MOV</code> , <code>ADD</code>).	Assembly language
High level	Designed to be easy for humans to read and write. Uses English-like words.	C++, Java, Python



The three levels of programming languages, with a sample line from each.

Common high-level languages

Language	Key idea	Used for
C++	Powerful and closer to hardware.	System software, operating systems, high-performance computer games.

Java	"Write once, run anywhere."	Large enterprise applications, Android mobile apps, server programming.
Python	Simple, readable syntax.	Data Science, Artificial Intelligence, Web Development, and learning to program.

WORKED EXAMPLE

Same task, three levels. Suppose you want to copy a value.

- Machine language: a pattern like 01010101 — you would need a chart to know what it means.
- Assembly: MOV AX, BX — "move the value in register BX into register AX". Shorter words, but you still need to know the hardware.
- Python: a = b — you simply say "a gets the value of b".

The higher the level, the less you need to know about the hardware and the faster you can write the program.

REMEMBER IT

"Low = Like the machine, High = Human." Mid level sits in the Middle and uses Mnemonics (both start with M).

COMMON MISTAKE

"Low level" does not mean "bad" or "weak". Low-level code runs very fast and controls hardware directly. It is just harder for humans to write.

اردو میں سمجھیں

پروگرامنگ زبانوں کی تین اقسام ہیں: لو لیول (Low Level) جو بائری 0 اور 1 میں ہوتی ہے، مڈ لیول (Mid Level) یعنی اسمبلی جو مختصر الفاظ (Mnemonics) استعمال کرتی ہے، اور ہائی لیول (High Level) جیسے پائتھن، جاوا اور سی پلس پلس جو انگریزی جیسے الفاظ استعمال کرتی ہیں اور انسانوں کے لیے آسان ہیں۔

CHECK YOURSELF

1. Which level is `MOV AX, BX`?
2. Which high-level language is famous for "write once, run anywhere"?
3. Give one use of C++.

Answers: 1. Mid level (assembly language). 2. Java. 3. Any one: system software, operating systems, high-performance games.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A program is:

- a. A set of instructions
- b. A mouse
- c. A cable
- d. A pie

MCQ 2. High-level languages are:

- a. Closer to English (Python, C++)
- b. Only 0 and 1
- c. Only analog
- d. Only SQL

MCQ 3. Machine language is:

- a. Binary codes for the CPU
- b. Python
- c. Urdu
- d. HTML always

MCQ 4. C++ is typically:

- a. High-level, closer to hardware than Python
- b. A DBMS
- c. A chart
- d. An OSI layer

MCQ 5. Python is popular in XI because:

- a. Simple syntax, huge libraries
- b. It is binary only
- c. It cannot print
- d. It is analog

MCQ 6. The translator of Python is a:

- a. Interpreter (also compiles to bytecode)
- b. Router
- c. Switch
- d. K-map

Extra short questions

1. Define program and programming language.
2. Differentiate high-level and low-level.
3. Name three high-level languages and one use each.
4. Why are computers "dumb" without programs?

Extra long question

1. Compare machine, assembly and high-level languages in a table (readability, speed, portability, example). Link why this course uses Python.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Program: instructions for a computer. Language: the notation used to write them.
2. High-level: English-like, portable. Low-level: close to hardware, faster but harder.
3. Python data/scripts; C++ systems/games; Java apps — (accept similar).
4. They only follow instructions; they do not understand the problem.

Long

1. Machine 0/1 fastest least portable; assembly mnemonics; high-level portable. Python: readable, batteries included, matches the Sindh practicals.

3.2 Career Growth with Python

LEARN IT

Python is one of the most in-demand languages in the world. Learning it opens many career paths:

- **Software Development:** build desktop and web applications.
- **Data Science and Data Analysis:** analyse data, create reports and make predictions.
- **Artificial Intelligence and Machine Learning:** develop intelligent systems, chatbots and recommendation algorithms (like "videos you may like").
- **Web Development:** use frameworks such as **Django** and **Flask** to build websites.
- **Automation and Scripting:** automate repetitive office or lab tasks (renaming 500 files, filling spreadsheets).
- **Cybersecurity and Networking:** write security tools and analyse network data.

Why Python? Python lets you focus on **problem-solving and logic** instead of memorising complex syntax rules. Compare printing a message in Python with C++:

```
# Python
print("Hello")
```

```
// C++
#include <iostream>
int main() {
    std::cout << "Hello";
    return 0;
}
```

WORKED EXAMPLE

Automation example. A school clerk must write "Roll No. 1" to "Roll No. 5" on a list. Python does it in two lines:

```
for n in range(1, 6):
    print("Roll No.", n)
```

OUTPUT

```
Roll No. 1
Roll No. 2
Roll No. 3
Roll No. 4
Roll No. 5
```

Change `6` to `501` and it writes 500 lines just as fast. (You will learn loops in 3.5.3.)

REMEMBER IT

Six careers — remember "**S-D-A-W-A-C**": **S**oftware, **D**ata, **A**I, **W**eb, **A**utomation, **C**ybersecurity.

COMMON MISTAKE

Django and Flask are **web frameworks** (for websites), not AI tools. AI/ML uses libraries such as TensorFlow and PyTorch (see 3.6.2).

اردو میں سمجھیں

پائٹھن سیکھ کر آپ سافٹ ویئر ڈیولپمنٹ، ڈیٹا سائنس، مصنوعی ذہانت (AI)، ویب ڈیولپمنٹ، آٹومیشن اور سائبر سیکیورٹی جیسے شعبوں میں کیریئر بنا سکتے ہیں۔ پائٹھن کا آسان انداز آپ کو مشکل اصول یاد کرنے کے بجائے مسئلہ حل کرنے پر توجہ دینے دیتا ہے۔

CHECK YOURSELF

1. Name two Python web frameworks.
2. Which career area builds chatbots and recommendation systems?

Answers: 1. Django and Flask. 2. Artificial Intelligence and Machine Learning.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Python careers include:

- a. Data analyst, web, automation, AI
- b. Only plumber
- c. Only analog technician
- d. Only OS kernel in 1950

MCQ 2. Python is used in AI because:

- a. Libraries like TensorFlow/Pandas exist
- b. It is binary only
- c. It cannot loop
- d. It has no syntax

MCQ 3. A scripting job often means:

- a. Automating repetitive tasks with Python
- b. Soldering NAND
- c. Drawing OSI only
- d. Figma only

MCQ 4. Learning Python in XI helps later with:

- a. XII data analysis and file handling
- b. Cooking only
- c. Cabling only
- d. Waterfall contracts only

MCQ 5. Syntax of Python is:

- a. Clear and beginner-friendly
- b. Harder than machine code
- c. Only umlauts
- d. Mandatory semicolons always

MCQ 6. Open-source means:

- a. The language/tools are free to use and inspect
- b. You must pay per loop
- c. Closed NAND
- d. Secret OSI

Extra short questions

1. Give four career paths that use Python.
2. Why do employers like Python?
3. What should an XI student build as proof?
4. Is Python the only language you will ever need?

Extra long question

1. Write a 10-mark "career connection" answer: three jobs, the Python skill each needs from this chapter, and one ethical warning.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Developer, data analyst, automation/scripting, AI/ML, teaching, testing.
2. Fast to write, huge ecosystem, readable by teams.
3. Small projects: bill calculator, file grader, attendance list.
4. No — but it is a strong first language and used in this syllabus.

Long

1. Analyst: data types+libraries. Developer: control structures. Automation: files later in XII. Ethics: do not scrape private student data without permission.

3.3 Integrated Development Environment (IDE) & VS Code

LEARN IT

An **IDE (Integrated Development Environment)** is a software tool that helps programmers **write, run and manage code in one place**. It is a complete workspace for programming. "Integrated" means many tools are joined together in one window.

Key features of an IDE:

- **Code Editor:** where you write and edit code.
- **Syntax Highlighting:** colours different parts of the code (keywords, strings, numbers) to make it easier to read.
- **Auto-completion:** suggests code while you type (type `pri` and it offers `print`).
- **Run/Execute Tool:** runs the program with one click.
- **Debugger:** helps you find and fix errors, for example by pausing the program at a chosen line.

WORKED EXAMPLE

Without vs with an IDE. Without an IDE you would write code in Notepad, save it, open a terminal and type `python file.py`, then go back to Notepad to fix mistakes. With an IDE you write, run (press the ► Run button) and debug in the same window, and errors are underlined while you type.

REMEMBER IT

IDE features — "**E-S-A-R-D**": Editor, Syntax highlighting, Auto-completion, Run, Debugger.

COMMON MISTAKE

An IDE does not write the program for you and does not catch logical mistakes. It only makes writing, running and debugging easier.

اردو میں سمجھیں

آئی ڈی ای (IDE) ایک ایسا سافٹ ویئر ہے جس میں کوڈ لکھنا، چلانا اور غلطیاں درست کرنا ایک ہی جگہ ممکن ہوتا ہے۔ اس کی اہم خصوصیات میں کوڈ ایڈیٹر، رنگین کوڈ (Syntax Highlighting)، خودکار تجاویز (Auto completion)، رن بٹن اور ڈیبیگر شامل ہیں۔

CHECK YOURSELF

1. What does "integrated" mean in IDE?
2. Which IDE feature suggests code while you type?

Answers: 1. Many programming tools (editor, runner, debugger) are combined in one program. 2. Auto-completion.

LEARN IT

VS Code is a **free, lightweight and powerful code editor** made by Microsoft and widely used for Python. By default it is **not a full IDE**, but it becomes one when you install **extensions** (add-ons), such as the **Python extension by Microsoft**.

VS Code interface elements

Element	Description
1. Top Menu Bar	Control over files, editing, navigation and workspace (File, Edit, Selection, View, Go, Run).
2. Side Bar	Contains the core tools: Explorer, Search, Source Control, Run & Debug and Extensions.
3. Explorer	Shows project files and folders. Helps you create, open and manage your Python files.
4. Search	Finds and replaces text in the open file or across the whole project.
5. Source Control	Connects with Git (a version control system) to track changes, undo mistakes and collaborate.
6. Run & Debug	Runs code, finds errors, pauses at specific lines and lets you inspect variables.
7. Extensions	Like an App Store for VS Code. Add features such as the Python extension, Prettier or Python Debugger.



A simplified VS Code window: menu bar, side bar, Explorer and the code editor.

WORKED EXAMPLE

Setting up and running your first program in VS Code:



```
print("Hello, World!")
```

OUTPUT

```
Hello, World!
```

The output appears in the **Terminal** panel at the bottom of the window.

REMEMBER IT

Extensions are the **"App Store"** of VS Code: just as apps give your phone new abilities, extensions give VS Code new abilities.

COMMON MISTAKE

Always save Python files with the `.py` extension (e.g. `hello.py`). Without it, VS Code does not know the file is Python, so there is no syntax highlighting and no Run button.

اردو میں سمجھیں

وی ایس کوڈ (VS Code) ایک مفت اور ہلکا پھلکا کوڈ ایڈیٹر ہے۔ یہ اپنے آپ میں مکمل آئی ڈی ای نہیں، لیکن ایکسٹینشنز (Extensions) لگانے سے آئی ڈی ای بن جاتا ہے۔ اس کے اہم حصے مینو بار، سائیڈ بار، ایکسپلورر، سرچ، سورس کنٹرول، رن اینڈ ڈیپگ اور ایکسٹینشنز ہیں۔

CHECK YOURSELF

1. Which VS Code element connects to Git?
2. Is VS Code a full IDE by default? How does it become one?
3. Where do you see and manage your project's files?

Answers: 1. Source Control. 2. No; it becomes an IDE by installing extensions (e.g. the Python extension). 3. Explorer.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. An IDE is:

- a. One app to write, run and debug code
- b. A printer
- c. A sort
- d. A cable

MCQ 2. VS Code is:

- a. A popular editor/IDE
- b. A DBMS
- c. An OSI layer
- d. A gate

MCQ 3. The terminal/run button is for:

- a. Executing the program
- b. Drawing pie charts only
- c. Sending TCP SYN only
- d. K-maps

MCQ 4. Syntax highlighting helps you:

- a. See keywords and mistakes
- b. Sort faster
- c. Encrypt disks
- d. Route packets

MCQ 5. A file explorer in VS Code shows:

- a. Project files
- b. OSI layer 1
- c. NAND groups
- d. Beachhead users

MCQ 6. Debugging in an IDE uses:

- a. Breakpoints and variable watch
- b. Only bubble sort
- c. Only pie charts
- d. Only Agile stickers

Extra short questions

1. Define IDE.
2. Name four VS Code areas.
3. Why not use Notepad only?
4. What is an extension?

Extra long question

1. Describe how you would create, run and debug a first Python file in VS Code. Include two interface elements from the notes.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Integrated Development Environment: editor + run + debug + files in one place.

2. Activity bar/sidebar (files), editor, terminal, status bar (and extensions).
3. No easy run, no highlighting, no debugger.
4. An add-on (Python plugin) that teaches VS Code a language.

Long

1. New file .py, type print, Run Python, read terminal. Mention top menu, explorer, terminal panel. Breakpoint on a wrong variable.

3.4 Fundamentals of Python Programming ★ Golden

LEARN IT

The basic structure of a Python program is simple and clear. A program takes data, **processes it using statements**, and finally **shows output**. Python runs the statements **top to bottom, one after another** (sequentially).

A **statement** is one complete instruction, usually written on one line, such as `print("Hi")` or `x = 5`.

1. Comments (#) — Comments are notes written in the code **for humans**. They start with the `#` symbol. The Python interpreter completely **ignores** them. Use them to explain what your code does.

2. Indentation (spaces) — C++ and Java use curly brackets `{ }` to mark a **block** of code (a group of statements that belong together). Python instead uses **indentation**: spaces at the beginning of a line. Indentation is **mandatory** in Python. The standard is **4 spaces** (or 1 tab).

```
# This is a comment. It explains the code below.
print("Hello, World!")
if 5 > 2:
    # The space before 'print' is the indentation
    print("Five is greater than two!")
```

OUTPUT

```
Hello, World!
Five is greater than two!
```

WORKED EXAMPLE

Reading the program line by line:

1. Line 1 starts with `#`, so Python skips it.
2. Line 2 prints `Hello, World!`.
3. Line 3 asks a question: is 5 greater than 2? The colon `:` says "a block follows".
4. Line 4 is a comment inside the block (ignored).
5. Line 5 is indented by 4 spaces, so it belongs to the `if` block. Because `5 > 2` is true, it runs.

What happens if you forget the indentation?

```
if 5 > 2:
print("Five is greater than two!")
```

OUTPUT

```
File "indent.py", line 2
    print("Five is greater than two!")
    ^
```

IndentationError: expected an indented block after 'if' statement on line 1

Python refuses to run the program and points to line 2. Add 4 spaces before `print` to fix it.

REMEMBER IT

Think of indentation like the points and sub-points in your notebook. Sub-points are written a little to the right to show they "belong" to the main point. A colon `:` at the end of a line always means "**indent the next line**".

COMMON MISTAKE

IndentationError? Check your spaces. **Mixing tabs and spaces** can break a Python program even when the code looks lined up. Choose 4 spaces and use them everywhere (VS Code inserts 4 spaces when you press Tab in a `.py` file).

اردو میں سمجھیں

پائنتھن پروگرام اوپر سے نیچے ترتیب سے چلتا ہے۔ `#` سے شروع ہونے والی لائن تبصرہ (Comment) ہوتی ہے جسے پائنتھن نظر انداز کر دیتا ہے۔ پائنتھن میں کوڈ کے بلاک کو بریکٹ کے بجائے انڈینٹیشن (Indentation) یعنی لائن کے شروع میں چار خالی جگہوں سے ظاہر کیا جاتا ہے، اور یہ لازمی ہے۔

EXAM TIP

Common questions: "What is the purpose of comments?", "Why is indentation important in Python?", "How is Python different from C++/Java in defining blocks?" Write: comments start with `#` and are ignored by the interpreter; Python uses indentation (4 spaces) instead of `{ }`; wrong indentation gives an `IndentationError`. Always give a 3–4 line code example.

CHECK YOURSELF

1. Predict the output:

```
# print("A")
print("B")
if 3 > 7:
    print("C")
print("D")
```

2. What is the standard amount of indentation in Python?

Answers: 1. B then D on separate lines (line 1 is a comment; `3 > 7` is false so C is skipped). 2. 4 spaces (or 1 tab).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Python runs statements:

- a. From top to bottom (unless control changes that)
- b. Randomly
- c. Only backwards
- d. Only on even lines

MCQ 2. A statement is:

- a. One instruction
- b. A mouse
- c. A layer
- d. A sprint

MCQ 3. Comments start with:

- a. #
- b. // always only
- c.